NEHRU COLLEGE OF ENGINEERING AND RESEARCH CENTRE
(AUTONOMOUS)
PAMPADY, THIRUVILWAMALA, THRISSUR DT., KERALA - 680588
NBA ACCREDITED PROGRAMMES - B.TECH CSE, B.TECH MTR | NAAC RE-ACCREDITED WITH 'A' GRADE
RECOGNIZED BY UGC UNDER SECTION(2F) OF THE UGC ACT, 1956 | ISO 9001:2015 CERTIFIED
APPROVED BY AICTE | AFFILIATED TO APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

MASTER OF COMPUTER APPLICATION
QUESTION BANK

SUBJECT: 24MCADST103 – Advanced Data Structures
SEM/YEAR: I/I
Subject Handled By: Dr Sudheer S Marar & Divya P

Module I

Review of basic data structures- Array, linked list and its variants, Stack, Queue and Trees. Set Data Structure: - Representation of sets, set implementation using bit string. Hashing: - Simple hash functions, Collision and Collision Resolution techniques, Disjoint sets- representations, Union, Find algorithms

Q No	Question
1.	Distinguish between linear data structure and non-linear data structure
2.	Differentiate between Stack and Queue.
3.	Difference between arrays and lists.
4.	What is a linked list? Specify the difference between singly, doubly and circular linked lists
5.	Distinguish graph and tree.
6.	What is meant by Hashing? Explain the different hash functions
7.	Explain any two collision resolution methods in Hashing
8.	Perform the Linear probing with the elements 8,26,37,51,11,36,42 with table size 10
9.	Perform the double hashing with the elements 7,16,33,53,15,17,26 with table size 10.
10.	What do you mean by collision and how can you handle it by using linear probing.
11.	What are the different collision resolution techniques in hashing? Explain any one of them.
12.	What is Set data structure? How is a Set implemented using Bit String?
13.	Define binary search tree
14.	What is Disjoint Sets? Explain with an example.
15.	How to represent a Set Data Structure?
16.	Explain Disjoint Set Data structure. What are the operations performed on Disjoint Set Data structure
17.	Explain real application of sets, hashing and disjoint sets

Module II

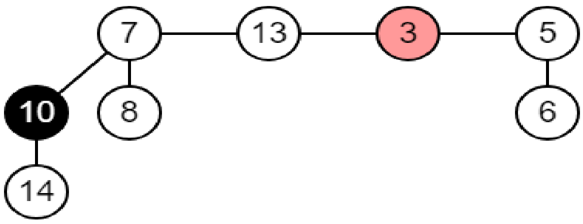
Advanced Tree Structures: - Balanced Binary Search trees, Red-Black trees- Properties of Red Black trees, Rotations, Insertion, Deletion. B-Trees- Basic operations on B-Trees – Insertion and Deletion, Introduction to Splay Trees and Suffix Trees

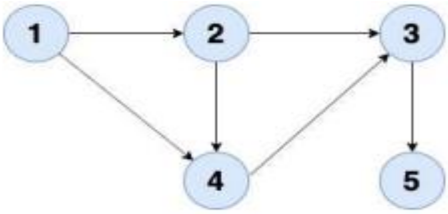
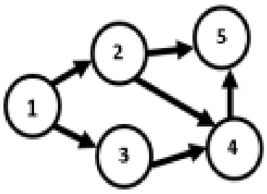
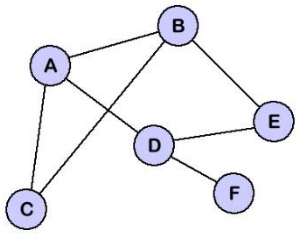
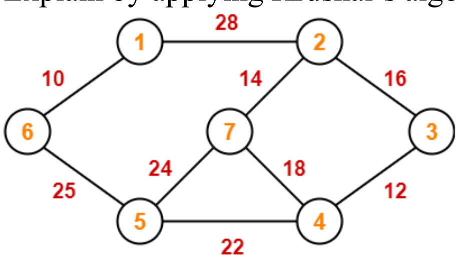
1.	Illustrate the properties of Red-Black Tree with example.
2.	Define Suffix Tree?
3.	Define Splay Tree. What are the advantages of Splay trees?
4.	Define Splay Tree. List the Rotations in Splay Tree
5.	Explain the characteristics of Balanced Binary Search Tree
6.	Explain Balanced Binary Search Tree. Give an example for a balanced binary search tree and an unbalanced one
7.	Define binary search tree (BST) and specify the steps showing the construction of a BST for the following data 10, 08, 15, 12, 13, 07, 09, 17, 20, 18, 04, 05

8.	Define binary tree and define the following binary trees with an example (i). Full binary tree (ii). Complete binary tree (iii) left & right skewed binary tree
9.	Give notes on B-Trees and Splay Trees
10.	Explain Red-Black tree insertion operations with examples
11.	Construct a red-black tree by inserting the keys in the following sequence into an initially empty red-black tree: 13, 10, 8, 3, 4 and 9. Show each step
12.	Describe BTree. How can we insert a key into a BTree?
13.	Define B tree. Explain B tree operations.
14.	Explain different cases of inserting nodes into a Red-Black Tree with an illustration.
15.	How a full node is splitted in B Tree Insertion procedure? Explain with a diagram.
16.	Explain how deletion is done in a Red Black Tree.
17.	Explain B-Tree insertion and Deletion operations with example.
18.	Define B-Tree. Specify its properties and describe the construction of a B-Tree for the following elements 5, 2, 13, 3, 45, 72, 4, 6, 9, 22.
19.	Create a red black tree by inserting following sequence of numbers 8,18,5,15,17,25,40 & 80
20.	Construct a B tree of order 3 by inserting numbers from 1 to 10
21.	Construct a Binary Search Tree by inserting the following sequence of numbers 10,12,5,4,20,8,7,15 and 13

Module III

Advanced Heap Structures: - Mergeable Heaps and operations on Mergeable Heaps. Binomial Heaps, Binomial Heap operations and Analysis, Fibonacci Heaps, Fibonacci Heap operations and Analysis.

1.	Write note on Fibonacci Heap
2.	Write note on mergeable heap?
3.	Explain Find-Min () operation of Binomial heap with example.
4.	Explain how the Union operation is performed on Binomial Heaps
5.	Describe how the Delete-Key operation is performed in a Fibonacci heap? Illustrate with an example.
6.	List out any three operations supported by a Mergeable Heap.
7.	Find the Potential of the Fibonacci Heap given below 
8.	List out the difference between min heap and max heap?
9.	Explain how the Decrease-Key operation is performed on Binomial Heaps. What is the Amortised Cost of this operation?
10.	Describe how Extract-Min operation is performed in a Fibonacci Heap? Illustrate with an example.
11.	Describe Binomial heap with example
12.	Explain Fibonacci Heap operations with example.
13.	A binomial heap has four binomial trees. Their degrees are 0,1,2 and 4. After you add an entry how many binomial trees will the heap have? What are the degrees of the trees?
14.	Explain the characteristics of Fibonacci Heap
15.	Draw a binomial heap whose keys are 6,3,5,18,1,10,7,9,16,10,20 Explain how union operation is performed in a Binomial heap
16.	1. Describe the Extract Min Operation in Fibonacci Heap with the help of an example.
17.	2. Discuss the advantage of using mergeable heap over binary heap?

	Module IV Advanced Graph Structures: Representation of graphs, Depth First and Breadth First Traversals, Topological Sorting, strongly connected Components and Biconnected Components Minimum Cost Spanning Tree algorithms- Prim’s Algorithm, Kruskal’ Algorithm, Shortest Path Finding algorithms – Dijikstra’s single source shortest paths algorithm
1.	Define Strongly Connected Components. Illustrate with an example
2.	Write a note on Minimum Costs Spanning Tree.
3.	What do you mean by Topological Sorting? Apply Topological Sorting to the given graph. 
4.	Discuss any one of the Topological Ordering of the graph. 
5.	Explain Topological Sorting. Apply Topological Sorting to the given graph. Describe each step. 
6.	Explain Depth First Search algorithm with an example
7.	Explain Breadth First Search algorithm with an example.
8.	Explain the Dijkstra’s Shortest Path algorithm with an example.
9.	Explain by applying Kruskal’s algorithm to find a minimum spanning tree of the following graph. 
10.	Explain Prims algorithm with an example
11.	Explain Kruskal’s algorithm with an example.
12.	Mention the two classic algorithms for the minimum spanning tree problem.
	Module V Blockchain Data Structure: - Blockchain Architecture, Blockchain Data Structures and Data types, Contract Data, Problems to be solved in Blockchain data analysis
1.	Explain on any four problems to be solved in Block chain Data Analysis.
2.	Explain Blockchain Architecture in detail.

3.	Describe the data types in Blockchain.
4.	Write notes on block chain data structure and its real word application
5.	Define Contract Data?
6.	Define smart contract.
7.	Explain Transaction model in Block Chain Technology.
8.	Explain the problems and challenges in Blockchain data
9.	Define Blockchain. Explain with the help of an example.
10.	Define blockchain technology and why might it be a catalyst for change for the financial sector
11.	Write a note on Merkle tree
12.	Explain Transaction in Blockchain with all the details.
13.	Explain Permissioned model of Block chain
14.	List out the advantages and disadvantages of Blockchain?
15.	Explain the role of header and body in a block chain block